

Rounding and Ordering on the Number Line

Answer Key

Grade 3 Math

Quick Answers

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|------------------|--|
| 1. 50 | 7. 600 |
| 2. 80 | 8. 733, 730, 703 |
| 3. 350 | 9. 700 |
| 4. 400 | 10. 600 |
| 5. 128, 218, 281 | 11. least to greatest = 358, 385, 835, — |
| 6. 500 | 12. — |
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What is verified

10 of 12 problems fully machine-checked against SymPy.
 — marks an answer only you can judge — problems 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 Math

2.NBT.A – problems 5, 8, 11

3.NBT.A.1 – problems 1, 2, 3, 4, 6, 7, 9, 10, 12

Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

1. *If they got 40: rounded down although the ones digit is 7*
 2. *If they got 90: rounded up although the ones digit is only 2*
 3. *If they got 340: rounded down although the ones digit is 7*
 4. *If they got 300: rounded down, looking at the ones digit instead of the tens digit*
 6. *If they got 400: rounded down, ignoring that 5 tens round up*
 7. *If they got 610: rounded up because the number felt big*
 9. *If they got 690: refused to cross into the next hundred and wrote 690*
 10. *If they got 700: rounded up because the ones digit is 9, instead of reading the tens digit*
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1. Round 47 to the nearest ten.

Solution: 47 sits between 40 and 50. The ones digit is 7, past the halfway mark at 45, so 47 is closer to 50.

50

2. Round 82 to the nearest ten.

Solution: 82 sits between 80 and 90, and its ones digit is only 2 — well before the halfway mark. It stays down at 80

3. Round 347 to the nearest ten.

Solution: 347 lies between 340 and 350. The ones digit 7 is past halfway (345), so round up. 350

4. Round 362 to the nearest hundred.

Solution: 362 lies between 300 and 400. For the nearest hundred the *tens* digit decides: 6 tens is past the halfway mark at 350, so round up. 400

5. Order least to greatest: 218, 128, 281.

Solution: Compare hundreds: 128 has 1 hundred, so it is least. 218 and 281 both have 2 hundreds, so the tens decide: 218 has 1 ten and 281 has 8 tens, so 218 comes first. On the line: 128, then 218, then 281. 128, 218, 281

6. Round 455 to the nearest hundred.

Solution: 455 lies between 400 and 500. The tens digit is 5, so 455 sits *past* the halfway mark at 450 — and halfway or more rounds up. 500

7. Round 604 to the nearest ten.

Solution: 604 lies between 600 and 610, and its ones digit 4 is before the halfway mark at 605. It rounds down — a big number can still round down! 600

8. Order greatest to least: 730, 703, 733.

Solution: All have 7 hundreds. Tens: 733 and 730 have 3 tens, 703 has none, so 703 is least. Ones break the 733/730 tie: 3 beats 0. 733, 730, 703

9. Round 695 to the nearest ten.

Solution: 695 lies between 690 and 700, exactly on the halfway mark — and halfway rounds *up*, even when that crosses into the next hundred. 700

10. Round 649 to the nearest hundred.

Solution: 649 lies between 600 and 700. The *tens* digit decides: 4 tens is before the halfway mark at 650, so 649 rounds down — the 9 ones cannot outvote the tens digit. 600

11. Mark 385, 358, 835 on the line and order them.

Solution: 358 and 385 both have 3 hundreds (ones: 358 has 5 tens, 385 has 8 tens, so 358 is smaller); 835 has 8 hundreds and sits far to the right. 358, 385, 835

Marks: 358 and 385 close together on the left part of the line, 835 far to the right — spacing should show 835 is much larger.

12. Mystery number: rounds to 350 (nearest ten) and 300 (nearest hundred).

Solution (instructor-judged): Rounding to the nearest ten gives 350 for the numbers 345 through 354. Rounding to the nearest hundred gives 300 only for numbers up to 349. Both clues together leave *345, 346, 347, 348, or 349* — accept any one of them. A good explanation marks the 345–354 stretch on the line and cuts it at 349 with the hundred clue.